Information Acquisition and the Value of Public Information in Network Games

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About me

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Information economics, economic networks, and monetary policy.

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1. Introduction

Incomplete information, networks, and welfare

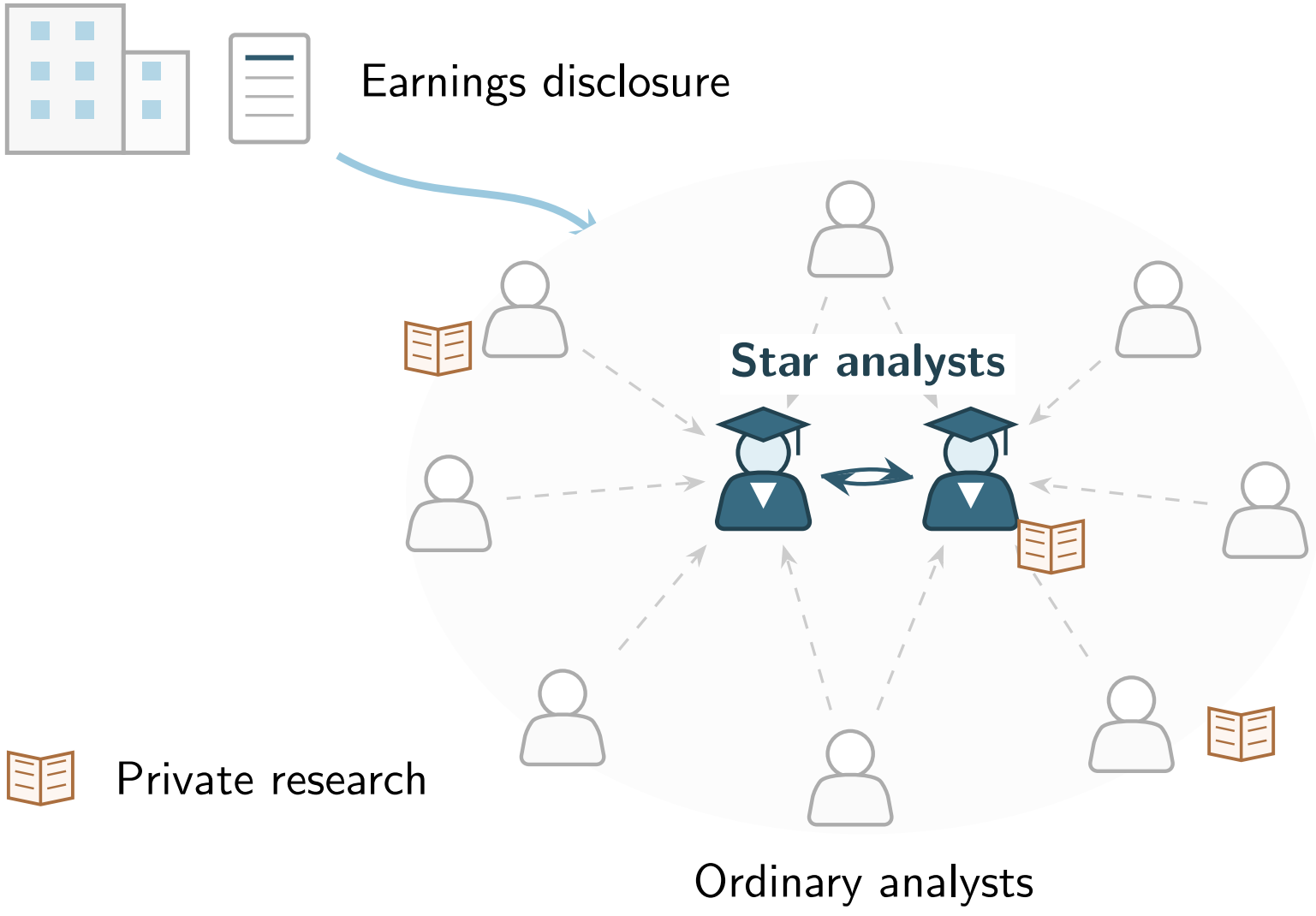
Information and strategic interactions

- **Decision makers routinely act with incomplete information.**
  - **Public information:** earnings reports, economic statistics, etc.
  - **Private information:** in-house research, proprietary data, etc.

- **Networks describe the strategic environment of decisions.**
  - Production and supply chains, technology adoption.
  - Agents differ in their positions and influences.

How does network structure shape private information acquisition and the private and social values of public information, and to what extent do private incentives align with social welfare?

An analyst example



This paper

We study the welfare effects of public information in a network game with endogenous private information acquisition.

- **Centrality amplifies crowding out.**
  - More central agents acquire less precise private information and reduce acquisition more sharply as public information improves.
- **Information acquisition can reverse welfare effects.**
  - Endogenous acquisition lowers the value of public information, especially where centrality and influence coincide in the same position.
- **Joint efficiency rules out welfare losses.**
  - Correcting the inefficiency in equilibrium acquisition alone does not guarantee a positive social value of public information.

Literature and contribution

- **Public information and welfare**
  - We study how network structure shapes the social value of public information with endogenous private information acquisition (Morris and Shin, 2002; Angeletos and Pavan, 2007).

- **Incomplete information in network games**
  - We connect asymmetric coordination motives to the private and social value of public information (Myatt and Wallace, 2019; Leister, 2020).

- **Efficiency and targeted interventions**
  - We implement the two social optima with action taxes and acquisition subsidies (Angeletos and Pavan, 2009; Galeotti et al., 2020).

Outline

Section	Takeaway
Model	Coordination in a directed network, with costly private information and a common public signal.
Equilibrium	Centrality amplifies crowding out (Prop. 1). Information acquisition can reverse the welfare effects of public information (Prop. 2).
Efficiency	Joint efficiency rules out welfare losses (Prop. 3). Action taxes and acquisition subsidies implement the two socially optimal benchmarks.

2. Model

Coordination incentives and costly private information

Preferences and network interactions

Each of $n \geq 2$ agents chooses $a_i \in \mathbb{R}$; θ is the unknown fundamental.

$$u_i(\mathbf{a}, \theta; \boldsymbol{\alpha}, \mathbf{G}) = -(1-\alpha_i) \underbrace{(a_i - \theta)^2}_{\text{adaptation}} - \alpha_i \sum_{j \neq i} g_{ij} \underbrace{(a_i - a_j)^2}_{\text{coordination}}.$$

- $0 \leq \alpha_i < 1$ measures aggregate coordination motives; some $\alpha_i > 0$.
- $\mathbf{G} = [g_{ij}]$ is directed and weighted.
  - $g_{ij} \geq 0$, $g_{ii} = 0$, and $\sum_j g_{ij} = 1$.
  - An arrow $i \to j$ means that i **wants to match** j.
- Effective coordination weights: $\widehat{g}_{ij} = \alpha_i g_{ij}$, collected in $\widehat{\mathbf{G}} = [\widehat{g}_{ij}]$.

Information structure

- θ draws from an improper uniform prior on $\mathbb{R}$.
- **Public information** is observed by everyone:

$$y = \theta + \varepsilon_y, \qquad \varepsilon_y \sim \mathcal{N}(0, 1/\tau_y).$$

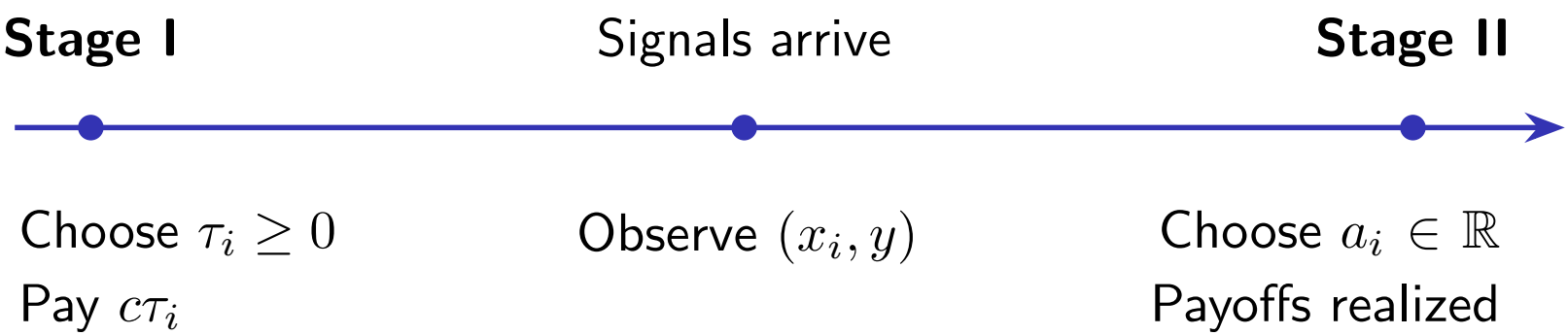
- **Private information** is observed only by agent i:

$$x_i = \theta + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, 1/\tau_i).$$

- Agent i chooses its precision $\tau_i \geq 0$ at a linear acquisition cost

$$C(\tau_i) = c\tau_i, \qquad c > 0.$$

Timing of the game



- Acquisition and private information are **unobserved by others**.
- At each stage, agents choose actions simultaneously.
- We study **linear perfect Bayesian equilibria**.

3. Equilibrium

Characterization and the value of public information

Equilibrium characterization

Result 1: Centrality amplifies crowding out.

Proposition 1

For sufficiently low c, there is a unique interior linear equilibrium:

$$a_i = \widehat{\beta}_i x_i + (1 - \widehat{\beta}_i) y,$$

where the weight on private information is

$$\widehat{\beta}_i = 1 - \tau_y \sqrt{c}\, \widehat{b}_i.$$

And the precision of private information is

$$\widehat{\tau}_i = \frac{1}{\sqrt{c}} - \tau_y \widehat{b}_i > 0.$$

Katz–Bonacich centrality: $\widehat{b}_i = [(\mathbf{I} - \widehat{\mathbf{G}})^{-1}\mathbf{1}]_i = \sum_{k=0}^{\infty} [\widehat{\mathbf{G}}^{\,k}\mathbf{1}]_i$.

- More central agents reduce private information acquisition more sharply when public information becomes more precise.

▸ Proof

The value of public information

Expected payoff and social welfare, net of acquisition costs:

$$U_i(\boldsymbol{\tau}, \tau_y) = \mathbb{E}[u_i] - c\tau_i, \qquad W(\boldsymbol{\tau}, \tau_y) = \sum_i U_i(\boldsymbol{\tau}, \tau_y).$$

Definition 1

The private value of public information for agent i is defined as

$$\frac{dU_i(\boldsymbol{\tau}, \tau_y)}{d\tau_y}.$$

The social value of public information is defined as

$$\frac{dW(\boldsymbol{\tau}, \tau_y)}{d\tau_y}.$$

Information acquisition and network position

Result 2: Information acquisition can reverse welfare effects.

Proposition 2

For sufficiently low c, endogenous information acquisition **reduces** *the private value of public information for each agent with $\alpha_i > 0$:*

$$\frac{\partial U_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_y} - \frac{dU_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} = c(\widehat{b}_i - 1) + O(c^{3/2}).$$

It also **reduces** *the social value of public information:*

$$\frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_y} - \frac{dW(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} = c \sum_i \widehat{d}_i^{in} \widehat{b}_i + O(c^{3/2}).$$

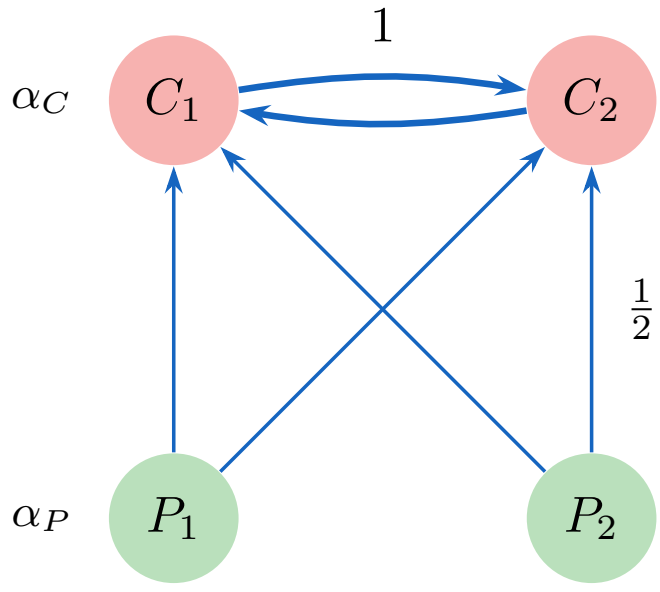
Weighted in-degree: $\widehat{d}_i^{in} = \sum_{j \neq i} \widehat{g}_{ji} = \sum_{j \neq i} \alpha_j g_{ji}$.

- The negative effect is prominent at central and influential positions.

▸ Decomposition ▸ Sign condition

Example: A core–periphery network

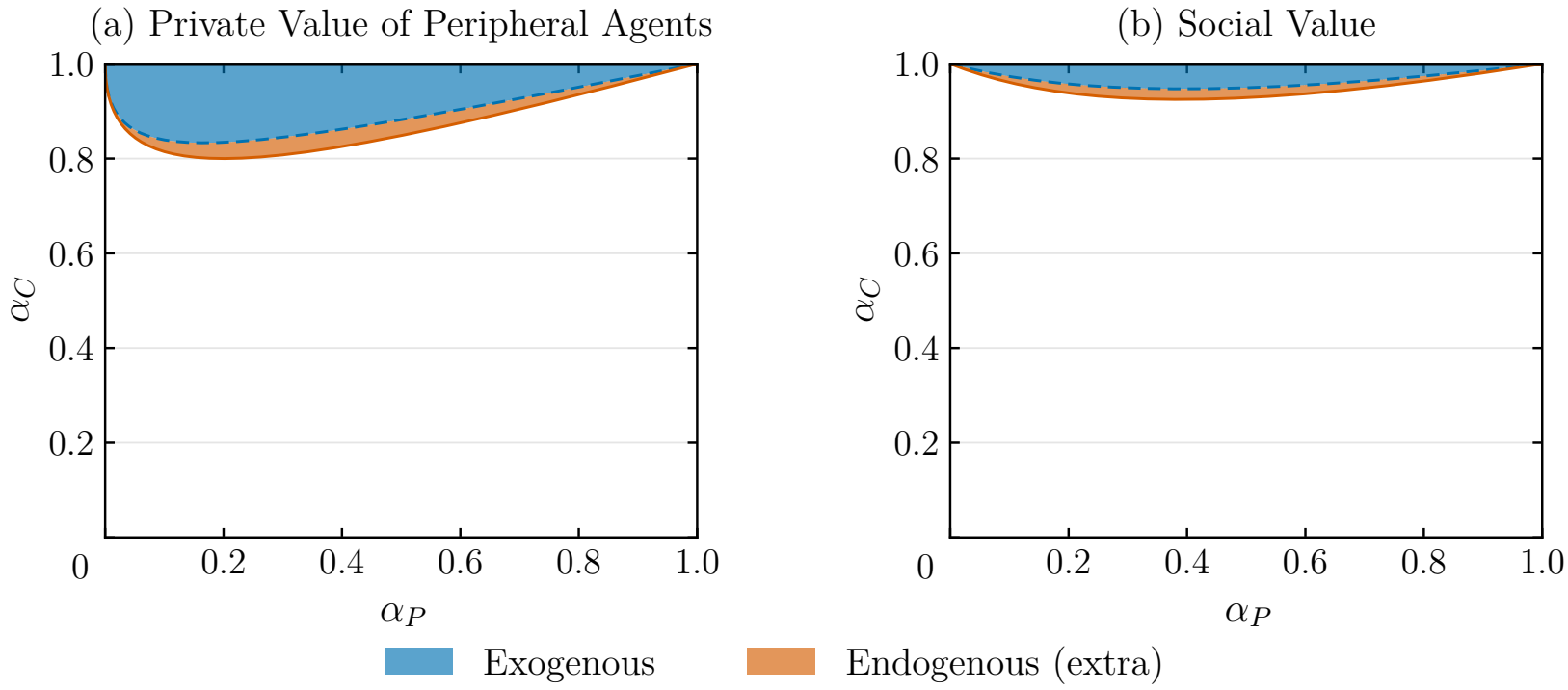
The coordination motives are α_C at the core and α_P at the periphery.



- Core agents follow each other; peripheral agents follow the core.
- When $\alpha_C > \alpha_P$, the core is both more central and more influential.

Centralities and influences

Welfare effects in the core–periphery example

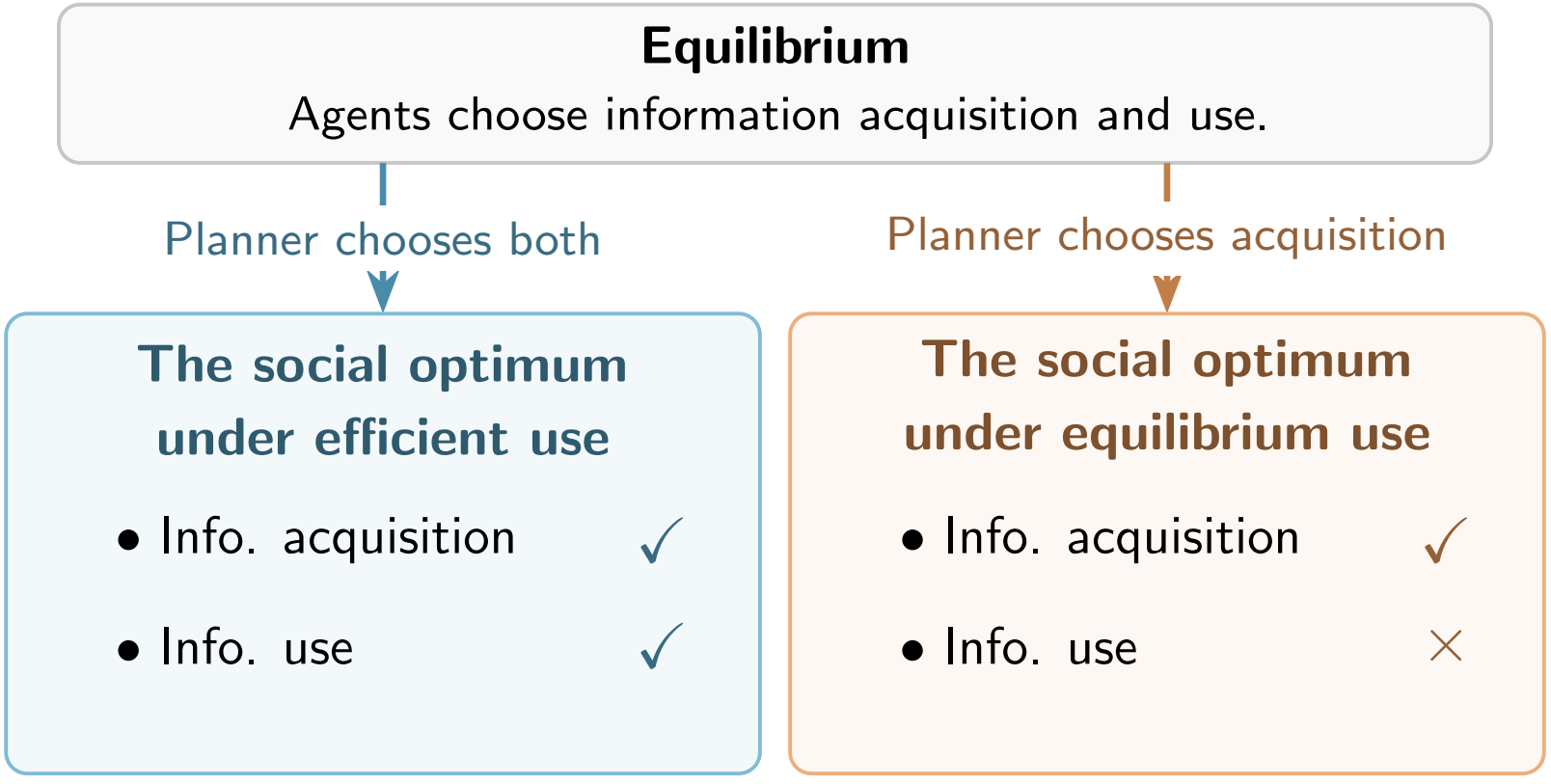


- Endogenous acquisition enlarges the regions in which public information harms peripheral agents and reduces social welfare.

4. Efficiency and Policy Implications

Two social optima and the instruments that implement them

Two socially optimal benchmarks



▸ Definitions ▸ Characterization

Welfare effects under socially optimal benchmarks

Result 3: Efficient acquisition alone is not enough.

Proposition 3

For sufficiently low c, the social value of public information is **positive** *at the social optimum under efficient use, while it is* **negative** *at the social optimum under equilibrium use, if and only if*

$$\sum_i \left[(1 - \widehat{d}_i^{in})(b_i^{**})^2 + \frac{2\widehat{d}_i^{in}}{\sqrt{1+\widehat{d}_i^{in}}} b_i^{**} \right] < 0.$$

Weighted Katz–Bonacich centrality

$$b_i^{**} = [(\mathbf{I} - \widehat{\mathbf{G}})^{-1}\mathbf{e}]_i, \qquad \mathbf{e} = (e_i)_{i \in N}, \quad e_i = (1 + \widehat{d}_i^{in})^{-1/2}.$$

- Network heterogeneity matters for the detrimental effect of public information at the social optimum under equilibrium use.

▶ Illustration

The crowding-in effect

Corollary 1

For sufficiently low c, at the social optimum under equilibrium use, the social value of public information is negative if and only if more precise public information increases total acquisition of private information:

$$\frac{dW(\boldsymbol{\tau}^{**}, \tau_y)}{d\tau_y} < 0 \quad \Longleftrightarrow \quad \frac{d}{d\tau_y} \sum_i \tau_i^{**} > 0.$$

When equilibrium information use places excessive weight on the public signal, the planner increases total private acquisition to mitigate the resulting distortion.

Policy implications

Action tax $\Rightarrow$ The social optimum under efficient use.

$$t_i^*(\mathbf{a}) = \sum_{j \neq i} \alpha_j g_{ji} (a_i - a_j)^2.$$

- Makes agents internalize coordination externalities on followers.
- More influence strengthens incentives to align with followers.

Acquisition subsidy $\Rightarrow$ The social optimum under equilibrium use.

$$s_i^{**} = \frac{\widehat{d}_i^{in}}{1 + \widehat{d}_i^{in}} + O(\sqrt{c}).$$

- Rewards the external benefits of private information acquisition.
- More influential agents receive higher subsidy rates.

▸ Action tax: Proposition 4 ▸ Subsidy: Proposition 5

5. Conclusion

What determines the welfare effects of disclosure?

Conclusion

1. More accurate public information changes the production of private information across the network.

2. Welfare losses are amplified when research is displaced at positions that are both central and influential.

3. Efficient acquisition can leave public information harmful; efficient acquisition and use restore a positive social value.

Main lesson: The welfare effect of public information depends on the incentives to produce and use information.

Thanks and comments welcome!

Appendix

Details for discussion

Equilibrium characterization

Proposition 1

For sufficiently low c, there is a unique interior linear equilibrium:

$$a_i = \widehat{\beta}_i x_i + (1 - \widehat{\beta}_i) y,$$

where the weight on private information is

$$\widehat{\beta}_i = 1 - \tau_y \sqrt{c}\, \widehat{b}_i.$$

And the precision of private information is

$$\widehat{\tau}_i = \frac{1}{\sqrt{c}} - \tau_y \widehat{b}_i > 0.$$

- $\widehat{b}_i \equiv [(\mathbf{I} - \widehat{\mathbf{G}})^{-1}\mathbf{1}]_i = \sum_{k=0}^{\infty} [\widehat{\mathbf{G}}^{\,k}\mathbf{1}]_i$ is the **Katz–Bonacich centrality** of agent i in the *equilibrium network* $\widehat{\mathbf{G}}$.

More central agents acquire less precise private information and place greater weight on the public signal when choosing their actions.

▸ Proof ◂ Back to Result 1

The private value of public information

Agent i's expected payoff net of acquisition costs is

$$U_i(\boldsymbol{\tau}, \tau_y) = \mathbb{E}[u_i] - c\tau_i.$$

Corollary 2(i)

For sufficiently low c, the private value of public information for agent i is negative if and only if

$$\frac{dU_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} < 0 \quad \Longleftrightarrow \quad \widehat{b}_i^2 < \sum_{j \neq i} \widehat{g}_{ij} \widehat{b}_j (\widehat{b}_j - 1).$$

More central neighbors rely more on the public signal. Following their actions can outweigh the benefit of a more accurate signal.

▸ Social value ◂ Back to Result 2

The social value of public information

Social welfare is defined as

$$W(\boldsymbol{\tau}, \tau_y) \equiv \sum_i U_i(\boldsymbol{\tau}, \tau_y).$$

Corollary 2(ii)

For sufficiently low c, the social value of public information is negative if and only if

$$\frac{dW(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} < 0 \quad \Longleftrightarrow \quad \sum_i \left[(1 - \widehat{d}_i^{in}) \widehat{b}_i^2 + \widehat{d}_i^{in} \widehat{b}_i \right] < 0.$$

- $\widehat{d}_i^{in} \equiv \sum_{j \neq i} \widehat{g}_{ji}$ is the **weighted in-degree** of agent i.

A detrimental effect requires strong reliance on the public signal to coincide with strong influence on others.

Private value | Back to Result 2

The role of endogenous acquisition

Proposition 2

For sufficiently low c, endogenous information acquisition **reduces** *the private value of public information for each agent with $\alpha_i > 0$:*

$$\frac{\partial U_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_y} - \frac{dU_i(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} = c(\widehat{b}_i - 1) + O(c^{3/2}).$$

It also **reduces** *the social value of public information:*

$$\frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_y} - \frac{dW(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} = c \sum_i \widehat{d}_i^{in} \widehat{b}_i + O(c^{3/2}).$$

Public information crowds out private information acquisition and the benefits it provides to other agents.

▸ Proof ◂ Back to Result 2

Welfare effects under socially optimal benchmarks

Proposition 3

For sufficiently low c, the social value of public information is **positive** *at the social optimum under efficient use, while it is* **negative** *at the social optimum under equilibrium use, if and only if*

$$\sum_i \left[(1 - \widehat{d}_i^{in})(b_i^{**})^2 + \frac{2\widehat{d}_i^{in}}{\sqrt{1+\widehat{d}_i^{in}}} b_i^{**} \right] < 0.$$

Weighted Katz–Bonacich centrality

$$b_i^{**} = [(\mathbf{I} - \widehat{\mathbf{G}})^{-1}\mathbf{e}]_i, \qquad \mathbf{e} = (e_i)_{i \in N}, \quad e_i = (1 + \widehat{d}_i^{in})^{-1/2}.$$

▸ Illustration ◂ Back to Result 3

Equilibrium concept

- $\sigma_i(x_i, y; \tau_i)$: action rules conditional on (x_i, y) and τ_i.

Definition 2

An allocation $(\widehat{\boldsymbol{\tau}}, \widehat{\boldsymbol{\sigma}})$ is an equilibrium if each agent i optimizes at both stages, given others' equilibrium strategies:

(i) At Stage II, for every acquired precision $\tau_i \geq 0$,

$$\widehat{\sigma}_i(x_i, y; \tau_i) \in \arg\max_{a_i \in \mathbb{R}} \mathbb{E}[u_i \mid x_i, y].$$

(ii) At Stage I, anticipating her optimal action rule,

$$\widehat{\tau}_i \in \arg\max_{\tau_i \geq 0} \left\{ \mathbb{E}[u_i] - c\tau_i \right\}.$$

Back to timing

Equilibrium: Acquisition and sequential optimality

Given others' rules, agent i's optimal use satisfies

$$\beta_i = \frac{\tau_i}{\tau_i + \tau_y} \left(1 - \alpha_i + \sum_j \widehat{g}_{ij} \beta_j \right).$$

Unobserved acquisition keeps others' rules fixed. The interior condition is

$$c = (\beta_i / \tau_i)^2.$$

Combining the conditions gives a contraction with modulus $\max_i \alpha_i < 1$:

$$\beta_i = \max \left\{ 0,\, 1 - \alpha_i - \tau_y \sqrt{c} + \sum_j \widehat{g}_{ij} \beta_j \right\}.$$

▸ Back to equilibrium

Interiority, information, and the comparison

- The equilibrium is interior if

$$0 < c < \left(\tau_y \max_i \widehat{b}_i\right)^{-2}.$$

- The prior is improper uniform; Gaussian signals give a proper posterior. Normalized linear rules yield losses independent of θ.
- Uniqueness is established within the class of linear equilibria.
- Welfare derivatives vary τ_y locally at fixed $(c, \boldsymbol{\alpha}, \mathbf{G})$, along the stated allocation.

Back to information

The welfare cost of displaced private information acquisition

Evaluated at equilibrium acquisition, with equilibrium use adjusting:

$$\frac{dW(\widehat{\boldsymbol{\tau}}, \tau_y)}{d\tau_y} = \frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_y} + \sum_i \frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_i} \frac{d\widehat{\tau}_i}{d\tau_y}.$$

The two ingredients are

$$\frac{\partial W(\widehat{\boldsymbol{\tau}}, \tau_y)}{\partial \tau_i} = c\, \widehat{d}_i^{in} + O(c^{3/2}), \qquad \frac{d\widehat{\tau}_i}{d\tau_y} = -\widehat{b}_i.$$

Their product aggregates the external benefit of research and the amount displaced at each position.

The private-value reduction is $c(\widehat{b}_i - 1) + O(c^{3/2})$ for $\alpha_i > 0$.

Back to Result 2

Payoff decomposition

Under $a_i = \beta_i x_i + (1-\beta_i) y$, the expected payoff of agent i is

$$\mathbb{E}[u_i] = -\underbrace{\left(\mathbb{E}[(a_i-\theta)^2] + \sum\nolimits_{j\neq i} \widehat{g}_{ij}\mathbb{E}[(a_j-\theta)^2]\right)}_{\text{prediction loss}}$$

$$+\underbrace{2\sum\nolimits_{j\neq i} \widehat{g}_{ij}\,\mathrm{Cov}(a_i-\theta, a_j-\theta)}_{\text{coordination gain}}.$$

Independent signal errors and $\tau_i, \tau_y > 0$ imply

$$\mathbb{E}[(a_i-\theta)^2] = \frac{\beta_i^2}{\tau_i} + \frac{(1-\beta_i)^2}{\tau_y},$$

$$\mathrm{Cov}(a_i-\theta, a_j-\theta) = \frac{(1-\beta_i)(1-\beta_j)}{\tau_y} \quad (j\neq i).$$

Holding all action coefficients fixed, the marginal gains are

$$\frac{\partial \mathbb{E}[u_i]}{\partial \tau_i} = \frac{\beta_i^2}{\tau_i^2} \geq 0, \qquad \frac{\partial \mathbb{E}[u_j]}{\partial \tau_i} = \widehat{g}_{ji}\frac{\beta_i^2}{\tau_i^2} \geq 0 \quad (j\neq i).$$

Both gains are strict when $\beta_i \neq 0$ and $\widehat{g}_{ji} > 0$.

◂ Back to Corollary 3

The core–periphery example: Centrality and influence

With two core and two peripheral agents:

$$\widehat{b}_C = \frac{1}{1-\alpha_C}, \qquad \widehat{b}_P = 1 + \frac{\alpha_P}{1-\alpha_C},$$

$$\widehat{d}_C^{in} = \alpha_C + \alpha_P, \qquad \widehat{d}_P^{in} = 0.$$

Each peripheral agent loses if and only if

$$(1 - \alpha_C + \alpha_P)^2 < \alpha_C \alpha_P.$$

Social welfare falls if and only if

$$(1 - \alpha_C + \alpha_P)^2 - \alpha_C \alpha_P < -(1 - \alpha_C^2).$$

Back to the network

The social optimum under efficient use

The planner jointly chooses the precision of private information and the action rules to maximize utilitarian welfare net of acquisition costs.

Definition 3

The socially optimal allocation under efficient use of information solves

$$(\boldsymbol{\tau}^*, \boldsymbol{\sigma}^*) \in \underset{\boldsymbol{\tau} \in \mathbb{R}^n_+,\, \boldsymbol{\sigma} \in \Sigma}{\arg\max} \left\{ \sum_{i \in N} \mathbb{E}[u_i] - c \sum_{i \in N} \tau_i \right\},$$

with expected payoffs evaluated at the efficient levels.

▸ Equilibrium use ◂ Allocation ◂ Back to the benchmarks

The social optimum under equilibrium use

The planner chooses the precision of private information; agents use information according to their equilibrium action rules $\boldsymbol{\sigma}^{\mathrm{eq}}(\mathbf{x}, y; \boldsymbol{\tau})$.

Definition 4

The socially optimal acquisition under equilibrium use of information solves

$$\boldsymbol{\tau}^{**} \in \arg\max_{\boldsymbol{\tau} \in \mathbb{R}^n_+} \left\{ \sum_{i \in N} \mathbb{E}[u_i] - c \sum_{i \in N} \tau_i \right\},$$

with expected payoffs evaluated under $\boldsymbol{\sigma}^{\mathrm{eq}}(\mathbf{x}, y; \boldsymbol{\tau})$, and

$$\boldsymbol{\sigma}^{**} \equiv \boldsymbol{\sigma}^{\mathrm{eq}}(\cdot, \cdot; \boldsymbol{\tau}^{**}).$$

▸ Efficient use ◂ Allocation ◂ Back to the benchmarks

The social optimum under efficient use: Allocation

The *efficient network* accounts for coordination in both directions:

$$\widetilde{\mathbf{G}} = (\widetilde{g}_{ij}), \qquad \widetilde{g}_{ij} = \frac{\widehat{g}_{ij} + \widehat{g}_{ji}}{1 + \widehat{d}_i^{in}}.$$

The **weighted Katz–Bonacich centrality** uses $\mathbf{e} = (e_i)_{i \in N}$:

$$b_i^* = [(\mathbf{I} - \widetilde{\mathbf{G}})^{-1}\mathbf{e}]_i, \qquad e_i = (1 + \widehat{d}_i^{in})^{-1/2}.$$

Proposition 6(i)

For sufficiently low c, the weight on private information and its precision are

$$\beta_i^* = 1 - \tau_y \sqrt{c}\, b_i^*, \qquad \tau_i^* = \sqrt{1 + \widehat{d}_i^{in}} \left(\frac{1}{\sqrt{c}} - \tau_y b_i^* \right).$$

The factor $1 + \widehat{d}_i^{in}$ includes the benefits of accuracy to followers.

▸ Equilibrium use ◂ Back to Definition 3

◂ Back to the benchmarks

The social optimum under equilibrium use: Allocation

The weighted Katz–Bonacich centrality uses the equilibrium network $\widehat{\mathbf{G}}$ and the vector $\mathbf{e} = (e_i)_{i \in N}$:

$$b_i^{**} = [(\mathbf{I} - \widehat{\mathbf{G}})^{-1}\mathbf{e}]_i.$$

Proposition 6(ii)

For sufficiently low c, the weight on private information and its precision have the expansions

$$\beta_i^{**} = 1 - \tau_y \sqrt{c}\, b_i^{**} + O(c), \qquad \tau_i^{**} = \sqrt{\frac{1 + \widehat{d}_i^{in}}{c}} + O(1).$$

Acquisition accounts for direct gains in accuracy and induced changes in equilibrium information use.

Efficient use | Back to Definition 4

Back to the benchmarks

Inefficiency in equilibrium acquisition

Corollary 3

For sufficiently low c, the equilibrium acquisition of private information is inefficiently low for agent i such that $\widehat{d}_i^{in} > 0$:

$$\widehat{\tau}_i < \tau_i^* \quad \textit{and} \quad \widehat{\tau}_i < \tau_i^{**}.$$

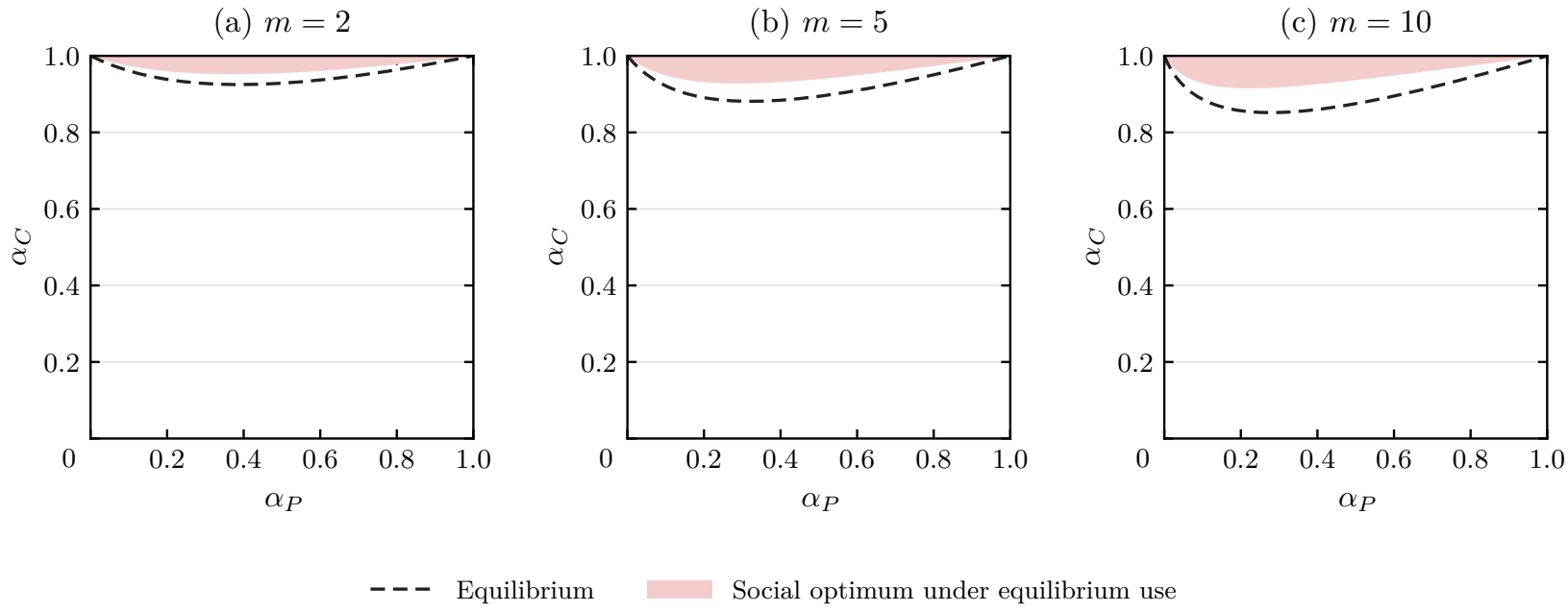
Total acquisition in equilibrium is also inefficiently low:

$$\sum_i \widehat{\tau}_i < \sum_i \tau_i^* \quad \textit{and} \quad \sum_i \widehat{\tau}_i < \sum_i \tau_i^{**}.$$

More accurate private information benefits the agent and her followers.

◂ Payoff decomposition ◂ Back to the benchmarks

Efficient acquisition can leave public information harmful



Back to Result 3

Policy interventions: action tax

- The planner taxes actions through $t_i^*(\mathbf{a})$, which depends on realized actions. Agent i's net payoff is

$$u_i - c\tau_i - t_i^*(\mathbf{a}) - \ell_i.$$

Here ℓ_i is a fixed transfer chosen before acquisition.

Proposition 4

For sufficiently low c, the socially optimal allocation under efficient use of information $(\boldsymbol{\tau}^*, \boldsymbol{\beta}^*)$ can be implemented by setting

$$t_i^*(\mathbf{a}) = \sum_{j \neq i} \alpha_j g_{ji} (a_i - a_j)^2.$$

The action tax corrects information use and also strengthens incentives for private information acquisition.

◂ Implementation ◂ Back to policy implications

Why the action tax also corrects acquisition

Under the tax $t_i^*(\mathbf{a}) = \sum_j \widehat{g}_{ji}(a_i - a_j)^2$,

$$\frac{\partial t_i^*}{\partial a_i} = 2\sum_j \widehat{g}_{ji}(a_i - a_j), \qquad \frac{1}{2}\frac{\partial^2 t_i^*}{\partial a_i^2} = \widehat{d}_i^{in}.$$

The first derivative makes agent i internalize losses to followers.

The second derivative changes the acquisition condition to

$$c = (1 + \widehat{d}_i^{in}) \left(\frac{\beta_i}{\tau_i}\right)^2,$$

which equals the condition at the social optimum under efficient use.

Back to the action tax

Policy interventions: acquisition subsidies

- The planner subsidizes private information acquisition at rate s_i. Agent i's net payoff is

$$u_i - c(1 - s_i)\tau_i - \ell_i.$$

Here ℓ_i is a fixed transfer chosen before acquisition.

Proposition 5

For sufficiently low c, the socially optimal allocation under equilibrium use of information $(\boldsymbol{\tau}^{**}, \boldsymbol{\beta}^{**})$ can be implemented by setting

$$s_i^{**} = 1 - \frac{1}{c}\left(\frac{\beta_i^{**}}{\tau_i^{**}}\right)^2 = \frac{\widehat{d}_i^{in}}{1 + \widehat{d}_i^{in}} + O(\sqrt{c}).$$

The implementing subsidy rates satisfy $s_i^{**} \in [0, 1)$, with $s_i^{**} > 0$ if and only if $\widehat{d}_i^{in} > 0$.

Subsidies internalize the external benefits of private information acquisition.

◂ Implementation ◂ Back to policy implications

Implementing the social optimum under equilibrium use

A subsidy changes the cost of acquisition to $c(1 - s_i)$ while leaving the incentives for subsequent information use unchanged.

At an interior acquisition choice, the first-order condition is

$$c(1 - s_i) = \left(\frac{\beta_i}{\tau_i}\right)^2.$$

Set the subsidy to match the private marginal benefit at the target allocation:

$$c(1 - s_i^{**}) = \left(\frac{\beta_i^{**}}{\tau_i^{**}}\right)^2.$$

The target allocation also satisfies equilibrium information use. Strictly concave acquisition payoffs and a contraction in the coefficients imply implementation as the unique linear equilibrium.

▸ Back to acquisition subsidies

The literature

- **An extensive literature on public information and welfare**
  - Seminal work includes Morris and Shin (2002).
  - Subsequent work on quadratic games with general externalities (Angeletos and Pavan, 2007), endogenous information choice (Colombo et al., 2014), and heterogeneous interactions (Myatt and Wallace, 2019; Murayama, 2026).

- **We ask:**
  - Which positions acquire more private information, and where does acquisition depart from socially optimal levels?
  - With endogenous acquisition, how does network structure affect the private and social values of public information?
  - Does efficient acquisition suffice to make more precise public information beneficial, or does efficient use also matter?

◂ Contributions ◂ Back to this paper

Incomplete Information in Decision-Making

- **Decision makers routinely act with incomplete information.**
  - Macroeconomic projections, equity valuation, and production planning.
- **Public information is important for decentralized decisions.**
  - Corporate earnings disclosure, official economic statistics, and industry-wide demand forecasts.
- **Private information is endogenously determined.**
  - In-house research, proprietary data purchases, and field surveys.

These information choices take place in settings where individuals' actions are strategically interdependent.

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Why Networks?

- **Networks describe the strategic environment of decisions.**
  - Cooperation and competition in production and supply chains: production planning and pricing.
  - Consensus in earnings forecasts: analysts' incentives to align their forecasts with those of their peers.

- **Agents differ in their positions of dependence and influence.**
  - Dominant firms and smaller rivals.
  - Prominent analysts and their less prominent peers.

Network heterogeneity shapes both incentives for private information acquisition and its spillover effects on others.

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